Types of Operating System & Unix/Linux

Types of OS

Single-user OS - supports only one user at a time. No multiple accounts, no concurrent users.

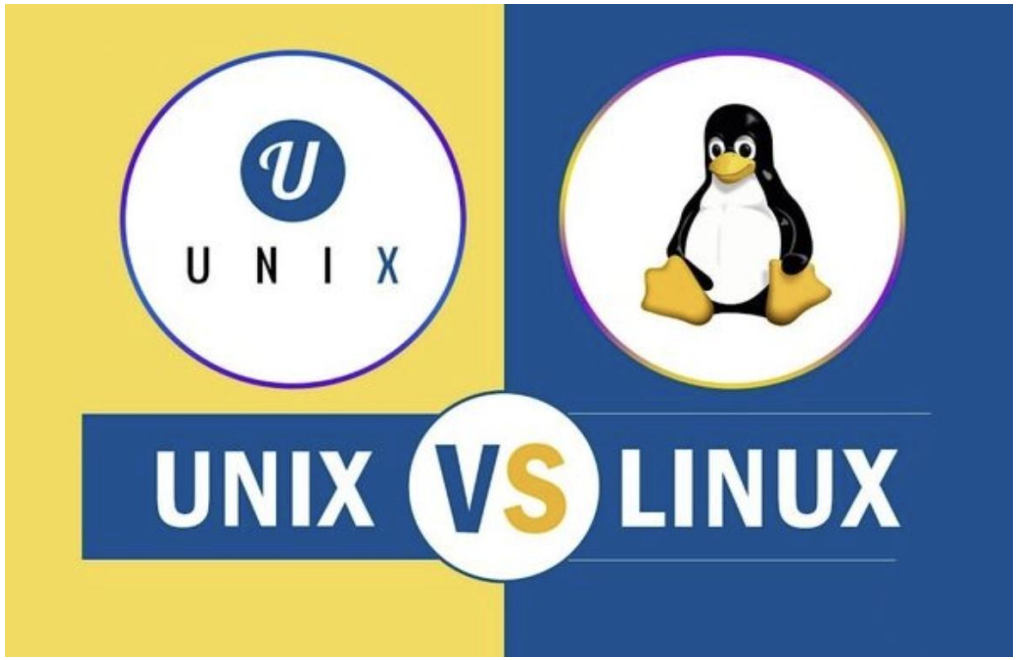
Multi-user OS - allows multiple users to use the computer simultaneously or remotely. Uses time-sharing, permissions, and user isolation.

Multitasking OS - can run multiple tasks/programs simultaneously. Uses scheduling and CPU time-slicing.

Distributed OS - manages a group of independent computers and makes them appear as one single system. Resources are shared across multiple machines in a network.

Real-time OS - an OS designed to process data and respond within a strict time constraint.Predictability is more important than raw speed.





Comparison of Unix and Linux

Unix

- Developed in 1969 at AT&T's Bell Labs by Ken Thompson & Dennis Ritchie.
- Proprietary (owned by companies like IBM, HP, Oracle).

Linux

- Created in 1991 by Linus Torvalds as a free alternative to Unix.
- Open-source and community-driven.

Usage

Unix: Mostly used in big enterprises, research, banking, and telecom systems.

Linux: Used everywhere — from servers, desktops, Android phones, supercomputers, IoT devices.



Hardware Support

Unix: Designed for specific hardware (IBM, HP machines).

Linux: Runs on almost any hardware (PCs, laptops, Raspberry Pi, supercomputers).